

Lab 02: PCB Design

Designing Schematics & PCB Layouts

Learning objectives

Upon successful completion of this lab, students should know how (to)...

- Use Electronic Design Automation (EDA) software;
- Describe circuits through schematic design;
- Design printed circuit board (PCB) layouts from a schematic;
- Produce drawings for both schematics and PCB layouts; and,
- Generate manufacturer files (Gerber, drill, etc.) for fabricators (fabs).

Overview

In this activity, you will use Electronic Design Automation (**EDA**) software to design a printed circuit board from a drawing of your choice; in completing this, you will gain experience in designing PCBs and an understanding of how hardware designs are created via schematics and schematics can be used to design boards for fabricators (JLCPCB, AISLER, etc.).

While modern PCB design can be extremely complex, the basic process is simple; even with a small working knowledge you can design boards for a variety of applications. The primary learning objectives of this lab are to understand the process of schematic capture, PCB layout, and fabrication, which will serve as a foundation for communication with other engineers working on cyber-physical systems and as a starting point for developing your own electronic assemblies.

A selection of schematics from typical computer engineering applications are available as part of this lab (see repo). Your goal is to recreate one of these schematics and design a simple PCB using the **KiCad** EDA software — a free and open-source cross-platform EDA software with a strong community engagement and support.

Instructions

This activity is broken into four steps: installing the software, creating the schematics and layout files, generating drawings, and completing a demonstration. The requirements for your implementation include:

- Accurate recreation of one of the provided schematics. Some guidelines to follow include:
 1. Be consistent with your use of through-hole or surface-mount components.
 2. Utilize net labels to make your design readable.
 3. Add notes to clarify any important information.
 4. Include your name, date, and other relevant information.
- Feasible implementation of a PCB design, with the following additional requirements:
 1. Use a layer stack up of two or four copper layers.
 2. Include at least one copper pour in your design.
 3. Include a board outline.
 4. Include appropriate ports (connectors or pin headers) for power, digital, and/or analog IO.
 5. Generate suitable fabrication outputs as if you were to place an order for your board.

KiCad Setup

Here is a [quickstart video](#) thanks to your TA Lucas! There are many useful online tutorials for KiCad (such as [this one](#)).

Download the [KiCad installer](#) for your platform and proceed as instructed to install. The default options should be suitable for most use-cases, which includes the built-in component libraries that you will need to find and use parts in your designs.

Submission

Your submission will be composed of the following elements:

- Project directory within a TGZ archive (kicad.tgz) on Gradescope,

```
kicad.tgz
├── PROJECT (dir)
│   ├── PROJECT.kicad_pcb
│   ├── PROJECT.kicad_pro
│   ├── PROJECT.kicad_sch
│   └── ...
```

- *Note:* We run KiCad's ERC and DRC against your project; both must come back with **no violations** (no errors, nor warnings). Violations caused by missing symbol or footprint libraries (i.e. not in default libraries) are ignored.

- Generated drawings of schematic and PCB layout in PDF format (one file), and
- Demonstration of PCB Design (in-person or recorded, see below).
 - *Note:* If you elect to submit a YouTube video, upload a plaintext file (video.txt) with just the URL.

Demonstration

Your demonstration will be completed either in person during office hours or as a link to an unlisted video resource (i.e. YouTube video). It is your choice how you convey this demo to us. Regardless of how you demo, it must include the following elements:

- Narrated overview of circuit schematic and PCB layout,
- Clear showcase of the board outline and copper pour(s),
- Clear showcase of the board passing the ERC (Electrical Rules Checker) & DRC (Design Rules Checker) (both errors and warnings),
- Clear showcase of how fabrication files are produced for the design.

Drawings

In addition to the project files, we require that you generate publish-ready drawing sheets for the schematic (must include original image of your chosen schematic) and the PCB layout.

These drawing sheets should:

- Self-explanatory in that purpose of components and the entire circuit is easily well understood (use text, if necessary),
- Metadata such as title and author in the drawing information section, and
- Be well-organized and presented.